



University of Central Arkansas
Department of Computer Science and Engineering
Fall 2026 – CSEC 3320 Computer Forensics

Instructor:

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Office Phone: 501-852-2050

Office: MCS 313

Course:

Title: COMPUTER FORENSICS

Course Number: 3320

CRN (Course Reference Number): 24718

Website: Brightspace

Lecture Times MWF – 11:00 AM-11:50 AM

Lecture Location: Room 328

Office Hours: Monday and Wednesday, 3:00–5:00 PM, or by appointment via email

Office hours are subject to change based on university committee responsibilities.

Course Description:

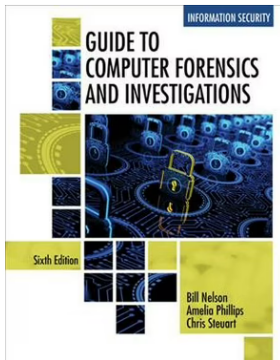
An introduction to digital forensics. Topics include creating and preserving digital evidence, data recovery and evidence collection algorithms, evidence construction and reconstruction, methods for certifying evidence, storing evidence, data acquisition, forensic analysis algorithms, image files, network forensics, logging methods to trace back attacks and digital trails, email investigations. A required course for Cybersecurity majors. Pre-requisites: CSEC 2300 INTRODUCTION TO CYBERSECURITY, CSCI 4305 LINUX/UNIX SYSTEMS

Course Objectives:

Upon successful completion of this course, you will be able to:

- Identify and classify sources of digital evidence in various computing environments.
- Preserve, acquire, and analyze digital evidence while maintaining integrity and chain of custody principles.
- Use standard digital forensic tools and techniques to examine and interpret evidence.
- Document, report, and explain findings from a digital forensic investigation in a clear and professional manner.

Textbook: Bill Nelson, Amelia Phillips, and Christopher Steuart. Guide to Computer Forensics and Investigations 6th Edition, 2018. ISBN: 9781337568944. Access Code is not required.



Required Software: Autopsy, FTK Imager, and other digital forensic tools as introduced during the course.

Tentative Topics and Schedule:

This schedule is tentative and may be adjusted during the semester based on class progress, instructional needs, or other circumstances. Any changes will be communicated to students in advance.

Week/Date	Topics	Assignment
Week 1 (Aug 21)	Classes Begin: Course Introduction Syllabus overview Introduction to digital forensics	
Week 2–3 (Aug 24 – Sep 4)	Module 1: Understanding the Digital Forensics Profession and Investigations (Chapter 1)	Quiz Assignment-1 (Released Sep 4)
	Digital Forensics Overview; Investigation Planning & Procedures; Forensic Tools & Workstations; Virtual Machines Autopsy Lab	
Week 4 (Sep 7–11)	Module 2: The Investigator’s Office and Laboratory (Chapter 2)	Quiz
	Lab Accreditation & Requirements; Lab Design &	Assignment-1 (Due Sep 16)

	Security; Forensic Workstations; Lab Planning	
Week 5-6 (Sep 14–25)	Module 3: Data Acquisition and Validation (Chapter 3)	Quiz
	Evidence Storage Formats; Acquisition Methods & Tools; Data Validation; RAID & Remote Acquisition FTK Imager Lab	Assignment-2 (Released Sep 25)
Week 7–8 (Sep 28 – Oct 9)	Module 4: Working with Windows and CLI Systems (Chapter 5)	Assignment-2 (Due Oct 7)
	File Systems; FAT/NTFS Structures; Windows Registry; Encryption; Startup Artifacts	Quiz Assignment-3 (Released Oct 9)
Week 9 (Oct 12–16)	Revision & Reading Midterm Exam (Oct 14) and Fall Break	Midterm Exam – Oct 14
Week 10-11 (Oct 19–30)	Module 5: Recovering Graphics Files (Chapter 8)	Assignment-3 (Due Oct 28)
	Graphics File Structures; Compression; File Signatures & Unknown Formats; Graphics Recovery Autopsy Lab	Quiz Assignment-4 (Released Oct 30)
Week 12–13 (Nov 2– Nov 13)	Module 6: E-mail and Social Media Investigations (Ch. 11)	Assignment 4 (Due Nov 11)
	E-mail Evidence & Headers; E-mail Servers & Artifacts; Social Media Evidence E-mail Analysis Lab	Quiz Assignment-5 (Released Nov 13)
Week 14-15 (Nov 16–23)	Module 7: Forensic Report Writing (Chapter 14) Writing Professional Digital Forensic Reports	Quiz; Final Exam will include a forensic report-writing question

Week 16 (Nov 30, Dec 2) (Dec 4: Study Day (No Class))	Course Review & Final Exam Preparation	Assignment-5 (Due Nov 30)
Dec 5 – Dec 10	Final Exam (date and time scheduled by the University): Dec 7, Time 2:00 PM- 4:00 PM	
December 14	Grades Due	

Summary

- **Several Quizzes**
- **5 assignments/labs**
- **Midterm**
- **Final exam**

Evaluation (%):

Assignments/Labs	30 %
Quizzes	25 %
Midterm	20 %
Final	25 %
TOTAL	100 %

A note about evaluation: Each question in the quizzes carries equal weight. Your overall quiz grade is calculated by dividing the total points earned across all quizzes by the total points available. Assignments follow a similar structure. Your assignment grade is calculated as the total points you earned divided by the total points available from all assignments.

Grading Scale:

A	90 - 100
B	80 - 89.99
C	70 - 79.99
D	60 - 69.99
F	0 - 59.99

Assignments: There will be several individual assignments/labs throughout the semester. To receive full credit, you must complete the assignments by the due dates. Late submissions will receive a 10% deduction from the assignment's total available points for each calendar day late. No credit will be given for submissions more than three calendar days late.

Valid reasons for missing a deadline: (1) School related functions (band, orchestra, sports events, etc.) (2) Medical reasons (illness, surgery, etc.) or accidents (e.g., motor vehicle) (3) Family emergency reasons (illness, surgery, death, etc.) in immediate family (parents, spouses, siblings, and children). In any case, prior contact with the instructor is required. Also, proper documentation (note from the coach, instructor, supervisor, physician, police, etc.) within three days is required. If your excuse is deemed to be valid, you will not be penalized for late submission for a maximum of three days.

Quizzes: There will be several quizzes during the semester. Some quizzes will be administered in class to assess comprehension and may also be used to document attendance. Students may attempt each online quiz up to three times. The highest score among all attempts will be recorded as the final score for that quiz

Important Dates

Please check the Academic Calendar at: <https://uca.edu/registrar/academic-calendar>.

Event	Date
Classes Begin	Aug 20 (First Class Aug 21)
Labor Day	Sep 7
Fall Break	Oct 15–18
Thanksgiving Break	Nov 25–29
Study Day	Dec 4
Final Exams	Dec 5–10
Grades Due	Dec 14

<https://uca.edu/registrar/files/2026/01/FINAL-EXAMINATIONS-2026-FALL-SEMESTER.pdf?1>

Office Hours:

Office hours are face-to-face. Zoom can be used for convenience. If you need to see me at another time please email me to set up the time. Links for Zoom will be sent.

Resources:

- Syllabus/Notes (Slides): Electronic copies will be made available on Brightspace.
- UCA Email: All email communication will be done via UCA email and Brightspace. You are required to check your email every day as information such as the availability of specifications, submission deadlines, and changes to the course schedule will be communicated via email.

Course Plan:

- The lectures will be in class.
- In case the classes need to be held online, ZOOM will be the environment, and the links will be posted in advance.
- I can be reached at the UCA e-mail address radhikari2@uca.edu.
- If you must miss class for a valid reason, please notify the instructor as early as possible. Students are responsible for making up missed course material and activities.

Classroom Conduct

The following rules are to ensure a classroom environment that is conducive to learning:

- Please turn off or mute any potentially disruptive electronic devices (pagers, mobile phones, portable media players, mobile computing devices, etc.)
- Please do not engage in private conversations during lectures.
- Please do not engage in non-class-related activities such as listening to music, browsing, playing games, etc.

Weekly Effort for the Class

On average, 6-10 hours of study per week will bring you to a good level of understanding of the concepts discussed in the classroom.

Academic Integrity Policy

The University of Central Arkansas affirms its commitment to academic integrity and expects all members of the university community to accept shared responsibility for maintaining academic integrity. Students in this course are subject to the provisions of the university's Academic Integrity Policy, approved by the Board of Trustees as Board Policy No. 709 on February 10, 2010, and published in the *Student Handbook*. Penalties for academic misconduct in this course may include a failing grade on an assignment, a failing grade in the course, or any other course-related sanction the instructor determines to be appropriate. Continued enrollment in this course affirms a student's acceptance of this university policy.

Americans with Disabilities Act Statement

The University of Central Arkansas adheres to the requirements of the Americans with Disabilities Act. If you need an accommodation under this Act due to a disability, please contact the Office of Accessibility Resources and Services (OARS), 450-3613.

Building Emergency Plan Statement

An Emergency Procedures Summary (EPS) for the building in which this class is held will be discussed during the first week of this course. EPS and Building Emergency Plan (BEP) documents for most buildings on campus are available at <https://uca.edu/go/bep-library>. Every student should be familiar with emergency procedures for any campus building in which he/she spends time for classes or other purposes.

The Title IX Disclosure

In furtherance of its core values— academic vitality, integrity, and diversity—UCA is dedicated to promoting a campus community free from discrimination. Title IX of the Education Amendments Act of 1972 requires all educational institutions to address gender-based discrimination on campus, and UCA implements these Federal requirements through a fair, consistent, and appropriate process of investigation and adjudication. Please see UCA's Title IX website (<https://uca.edu/titleix/>) for the university's policy, relevant forms, training opportunities, and related information.

Attendance and Drop Policy:

Regular class attendance is expected and strongly encouraged. Some in-class quizzes and activities may also serve as attendance records. Students who wish to withdraw from the course are responsible for following the University's official withdrawal procedures and deadlines.

Other Policies

Students should familiarize themselves with all policies included in the Student Handbook, particularly the following:

- Sexual Harassment Policy
- Academic Policies

AI Technologies

AI tools may be used as **learning aids** to help understand concepts and review material. However, students are responsible for the accuracy of their work and **must be able to explain, reproduce, and justify anything they submit**. The instructor may ask students to demonstrate their understanding of submitted work.

AI tools are not permitted during quizzes or exams unless explicitly authorized by the instructor. Unauthorized AI use may be treated as an academic integrity violation.

Student Evaluations

Student evaluations of a course and its professor are a crucial element in helping faculty achieve excellence in the classroom and the institution in demonstrating that students are gaining knowledge. Students may evaluate courses they are taking starting on the Monday of the thirteenth week of instruction through the end of finals week by logging in to myUCA and clicking on the Feedback Hub.

A final note:

If you have a concern about this class, please contact the instructor/me first. If you would rather talk to someone else, please contact the department chair (ecelebi@uca.edu).